Luke McGinnis

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EDUCATION

Eastern Michigan University – Ypsilanti, MI — Graduated May 2026
B.S in Mechanical Engineering, Minor in Math
GPA: 3.6/4.0, Dean's List, 5 semesters

Skyline High School – Ann Arbor, MI
Design Lead, Skyline Robotics Team, FIRST Robotics Competition

EXPERIENCE

Engineering Research Assistant, Eastern Michigan University — April 2025 – May 2026

- Designed and CNC machined torsion and tensile test benches using Autodesk Inventor and Inventor CAM, replacing low-rigidity stress testing equipment to enable sensor endurance testing for ongoing research.
- Co-authored IEEE 2026 publication on flexible strain gauge reliability by designing sensor paths in Python and KiCad, fabricating prototypes, and conducting stress tests to characterize print parameter effects on sensor endurance.

Mechanical Engineering Intern, SimCraft – Atlanta, GA — May 2025 – September 2025

- Reconstructed CAD for 3 axes of a 6-DOF driving simulator using Fusion 360, resolving design inconsistencies, and improving manufacturability.
- Developed a parametric simulator model to enable product variant configuration and support integration with an online sales configurator.

Intern, Mechanical Simulation – Ann Arbor, MI — June 2018 – August 2019

- Used RoadRunner (software now owned by Mathworks) to recreate 3D road maps for simulating autonomous driving systems. Evaluated and documented RoadRunner's potential for generating future road designs for ADAS and autonomous driving technologies.

PROJECTS

Senior Capstone Project: Designed and prototyped a parametric, EV battery pack in SolidWorks to improve accessability to vehicle electrification. Design included integrated thermal cooling plate, implementation of UL standards, and a manufacturability analysis.

Machine Design Project: Modeled and engineered a three-stage helical gearbox in SolidWorks, applied American Gear Manufacturers Association standards, and performed detailed shaft stress analyses to ensure an infinite fatigue life.

Fusion Reactor Model: Developed dynamical systems model of a hypothetical fusion reactor's stability based on feedback loop and numerical differential equation analysis of deuterium-tritium fusion burn reaction.

SKILLS

Design Tools: SolidWorks, NX, Autodesk Fusion 360, Autodesk Inventor, KiCad
Languages: Python (with pandas, NumPy, and TensorFlow), MATLAB, Java
Manufacturing: Inventor CAM, Fusion 360 CAM, 3-axis CNC machining, 3D Printing